$$\begin{array}{ccc} P_1 \times \mathrm{S}^0 & \xrightarrow{f} & P_0 \\ {\scriptstyle \pi'}\downarrow & & \downarrow \\ P_1 & \longrightarrow & \overline{P} \end{array} \qquad P_1 \underset{\partial^+}{\overset{\partial^-}{\rightrightarrows}} P_0 \longrightarrow \overline{P}$$